

SECTION 2.9

DIMENSION AND RANK

ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03)

Spring, 2018

This section continues the discussion of subspaces and bases for subspaces, beginning with the concept of a coordinate system.

COORDINATE SYSTEMS

The main reason for selecting a basis for a subspace H ; instead of merely a spanning set, is that each vector in H can be written in only one way as a linear combination of the basis vectors of H . To see why, suppose $\mathcal{B} = \{b_1, \dots, b_p\}$ is a basis for H , and suppose

$$x = c_1 b_1 + \dots + c_p b_p \quad \text{and} \quad x = d_1 b_1 + \dots + d_p b_p.$$

Then, subtracting gives

$$0 = x - x = (c_1 - d_1)b_1 + \dots + (c_p - d_p)b_p.$$

Since $\mathcal{B}$ is linearly independent, the weights $c_i - d_i$ must all be zero. That is, $c_i = d_i$ must hold for $1 \leq i \leq p$, which shows that the two representations of x given above are actually the same – i.e. a representation of a vector in a given basis is *unique*.

Definition 1. Suppose the set $\mathcal{B} = \{b_1, \dots, b_p\}$ is a basis for a subspace H . For each $x \in H$, the *coordinates* of x relative to the basis $\mathcal{B}$ are the weights $c_1, \dots, c_p$ such that

$$x = c_1 b_1 + \dots + c_p b_p$$

and the vector in $\mathbb{R}^p$ given by

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the *coordinate vector* of x relative to $\mathcal{B}$, or the $\mathcal{B}$ -*coordinate vector* of x .

In general, if $\mathcal{B} = \{b_1, \dots, b_p\}$ is a basis for H , then the mapping $x \mapsto [x]_{\mathcal{B}}$ one-to-one correspondence that makes H look and act the same as $\mathbb{R}^p$ (even though the vectors in H themselves may have more than p entries). It's straightforward to check that this correspondence between H and $\mathbb{R}^p$ is one-to-one (injective) and preserves linear combinations (i.e. $[u + v]_{\mathcal{B}} = [u]_{\mathcal{B}} + [v]_{\mathcal{B}}$). This is an important concept, which we will later use quite heavily. More specifically, we say that an injective, linear combination preserving correspondence between H and $\mathbb{R}^p$ is an *isomorphism*, and we say that H and $\mathbb{R}^p$ are *isomorphic*. In essence, the easiest way to think about this is as follows: if two vector spaces are isomorphic, then they're really the same vector space, but with the vectors labeled differently.

Example 1. Let $v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and let $x = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$. As v_1 is not a scalar multiple of v_2 , the set $\mathcal{B} = (v_1, v_2)$ is linearly independent. As such, $\mathcal{B}$ forms a basis for $H = \text{span}\{v_1, v_2\}$. So, observe that x is in H , as

$$2v_1 + 3v_2 = 2 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix} = x.$$

Knowing this, it's easy to find $[x]_{\mathcal{B}}$, the coordinate vector of x relative to $\mathcal{B}$, since $c_1 v_1 + c_2 v_2 = x$ for the constants $c_1 = 2$ and $c_2 = 3$. That is,

$$[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

THE DIMENSION OF A SUBSPACE

Definition 2. The *dimension* of a nonzero subspace H , denoted by $\dim(H)$, is the number of vectors in any basis for H . The dimension of the zero subspace $\{0\}$ is defined to be zero.

This tells us that $\mathbb{R}^n$ is n -dimensional, as every basis for $\mathbb{R}^n$ consists of n vectors. Similarly intuitive reasoning holds for subspaces in general: a plane through zero in $\mathbb{R}^n$ (for $n \geq 2$) is two dimensional, a line through zero in $\mathbb{R}^n$ (for $n \geq 1$) is one dimensional, etc.

Example 2. Let A be an $m \times n$ matrix and suppose $Ax = b$ has a solution for every $b \in \mathbb{R}^m$. Then, we know that the columns $a_1, \dots, a_n$ of A must span $\mathbb{R}^m$. So, it follows that $\dim(\text{Col}(A)) = n$ if $Ax = b$ has a solution for every $b \in \mathbb{R}^m$.

Definition 3. The *rank* of a matrix A , denoted by $\text{rank}(A)$, is the dimension of the column space of A . That is,

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

Seen another way, since the pivot columns of a matrix A form a basis for $\text{Col}(A)$, the rank of A is just the number of pivot columns in A .

Example 3. To determine the rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$$

we simply reduce A to echelon form:

$$\begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since this matrix has 3 pivots, A has 3 pivot columns. So, we may conclude that $\text{rank}(A) = 3$.

Since the number of pivot columns plus the number of non-pivot columns is exactly the number of columns, the dimensions of $\text{Col}(A)$ and $\text{Nul}(A)$ have the following useful connection:

Theorem 1. *If a matrix A has n columns, then $\text{rank}(A) + \dim(\text{Nul}(A)) = n$.*

The following theorem is important for applications and will be needed in Chapters 5 and 6. The theorem (which is proved in Section 4.5 of your book) is certainly plausible, if you think of a p -dimensional subspace as isomorphic to $\mathbb{R}^p$. The Invertible Matrix Theorem shows that p vectors in $\mathbb{R}^p$ are linearly independent if and only if they also span $\mathbb{R}^p$.

Theorem 2. *Let H be a p -dimensional subspace of $\mathbb{R}^n$. Then, any linearly independent set of exactly p elements in H is automatically a basis for H . Also, any set of p elements of H that spans H is automatically a basis for H .*

RANK AND THE INVERTIBLE MATRIX THEOREM

The various vector space concepts associated with a matrix provide several more statements for the Invertible Matrix Theorem.

Theorem 3. *Let A be an $n \times n$ matrix. Then, the following statements are equivalent:*

1. A is an invertible matrix
2. The columns of A form a basis for $\mathbb{R}^n$
3. $\text{Col}(A) = \mathbb{R}^n$
4. $\dim(\text{Col}(A)) = n$
5. $\text{rank}(A) = n$
6. $\text{Nul}(A) = \{0\}$
7. $\dim(\text{Nul}(A)) = 0$